

From a Methods Section to an Analytic Cohort
SPPH 604 · a worked NHANES reproduction

Kueh et al., BMJ Open 2026;16:e113719 · NHANES 2007–2018 + linked
mortality

A Methods sentence is not an analysis rule.

Someone has to decide what it means. That someone is you.

Today: one published paper, one public dataset, and every decision in between.

Start with four words from the Methods

*"The study inclusion criteria were participants **aged** **>18 years** with evidence of MASLD."*

— Kueh et al., BMJ Open 2026

Before you write any code: who exactly qualifies?

- ▶ **>18 means 19 and older**
- ▶ **≥ 18 means 18 and older**

There is no third reading. One of them is what the authors did, and the paper does not tell you which.

One character moves 70 people

70 participants are exactly 18

Age rule	Final N	Deaths
Published paper	6,300	585
Our reproduction, ≥ 18	6,371	586
Our reproduction, > 18	6,301	585

The strict reading — the one the paper literally prints — lands one row from the published N and reproduces the death count exactly.

But the evidence does not settle it

Evidence	Points to
The printed Methods text says >18	> 18
N = 6,301 and 585 deaths vs published 6,300 / 585	> 18
Published mean age 51.0 (SD 16.6); ours is 51.0 (16.6) under ≥ 18 , 51.4 (16.3) under >18	≥ 18

So we cannot resolve it — and saying so is the finding.

Report the ambiguity, show both, and let the reader see what turns on it.

Typography is evidence of what was printed, not proof of what was intended.

That was the whole method

Paper — what does the article actually say?

Problem — what is ambiguous, missing, or unmeasurable in the data?

Decision — what rule did we implement, and why?

Consequence — who entered or left, and what *kind* of change was it?

Every decision gets one of three marks:

-
- | | |
|-----|--|
| ✓ | direct translation — the paper determines the answer |
| ◆ | judgment call — the paper leaves it open, we chose |
| [!] | unresolved conflict — the evidence disagrees with itself |
-

What NHANES is, in one slide

Cycle — one two-year survey wave. We use six: 2007–08 through 2017–18.

Component — a topic file inside a cycle: demographics, body measurements, blood pressure, laboratory panels, questionnaires.

Variable — a column inside a component: age, waist circumference, triglycerides.

SEQN — the participant ID that joins components for the same person.

! Important

NHANES is a **repeated cross-sectional survey**. Each cycle samples new people. Mortality is added afterwards by linkage, which gives prospective follow-up — but never a second measurement of anybody.

The journey: 59,842 people to 6,371

59,842 → 36,580 → 6,371

59,842	everyone in six NHANES cycles
36,580	after age eligibility
6,371	after the disease definition, the exposure, and mortality linkage

The paper reports **6,300**.

Three numbers now. The reasons behind them are the next six slides — and the reasons matter more than the arithmetic.

Not everything that leaves is missing data

Triglycerides are measured only in NHANES's **fasting laboratory subsample**, which carries its own weight (WTSAF2YR).

What happened	How many adults	What kind of loss
Outside the fasting-weight frame	21,618	design
Inside it, but missing an input we need	792	item missingness

! Important

These need opposite responses. A design restriction is answered by the right weight and a sentence naming the population you are left with. Item nonresponse is answered by a missing-data analysis. Adding them together and calling the total “attrition” is wrong three ways at once.

A code means what *its own* codebook says

Triglycerides (LBXTR)

77 is a triglyceride of **77 mg/dL**.

A real measurement.

Drinks per drinking day (ALQ130)

777 means **refused**. 999 means **don't know**.

Not measurements at all.

 Warning

Apply one global missing-value rule and you either delete valid laboratory values or convert **9 refusals** into people drinking 777 drinks a day. There is no NHANES-wide rule. The codebook for *that variable*, in *that cycle*, decides.

Defining the disease takes four measurements

MASLD = fatty liver + at least one cardiometabolic feature. No liver-imaging measure spans all six cycles (2017–18 added elastography, the earlier five have none), so the paper uses a **Fatty Liver Index** built from four routine measurements:

triglycerides · **BMI** · liver enzyme (GGT) · **waist circumference**

Threshold: $FLI \geq 60$. That leaves **6,376** people with fatty liver.



◆ Notice what just happened

The exposure in this study is **BMI crossed with waist-to-height ratio**. Two of the four inputs to the *disease* definition are BMI and waist. **The disease definition and the exposure share their ingredients** — keep that in mind for M1.

A criterion can sound decisive and do nothing

The paper requires fatty liver **plus at least one** of five cardiometabolic criteria.

It excluded 2 people

6,376 with fatty liver → **6,374** with MASLD.

Why. The first criterion is essentially “high BMI or high waist”, and FLI is *built from* BMI and waist — so selecting on $FLI \geq 60$ heavily **enriches** the cohort for that criterion. It does not guarantee it: **89** people here have $FLI \geq 60$ with a BMI under 25, and 42 fail criterion (i) outright.

Report a filter by its effect, not its definition. “MASLD required at least one cardiometabolic criterion” sounds like work; here it removed two people.

[!] A vague exclusion has to be made precise

“Individuals with significant alcohol consumption — more than two drinks per day for men and more than one for women — were excluded.”

Problem. NHANES has no “drinks per day” variable. It has *drinks per drinking day* and, separately, *how often you drink*. Three drinks every Saturday is $ALQ130 = 3$ — but **0.4 drinks/day**.

Decision. Rebuild average intake as frequency $\times$ quantity. That flags **427 people**.

Consequence. Applying it takes us to **5,944** — *further* from the paper’s 6,300 than leaving it off (6,371). So we build it, ship it, log the setting, and do not apply it.

The exposure is a 2×2 you have to build

Obesity = $\text{BMI} \geq 30$ (≥ 25 for Asian participants). High central adiposity = waist-to-height ratio ≥ 0.6 .

	WHtR < 0.6	WHtR $\geq$ 0.6
Obesity	Group I — 391	Group II — 4,585
No obesity	Group III — 787	Group IV — 608

💡 ✓ How we know the construction is right

Our four group BMI medians are **30.9 / 35.2 / 27.6 / 28.8** — identical to all four published values. That check is independent of the N disagreement, so it isolates the group logic from everything else.

[!] The variable you need may not exist in every cycle

Why we care. Asian participants get a different BMI cut-point (25, not 30).

Problem. NHANES did not identify Asian participants separately until **2011–2012**. For 2007–2010 they are inside a combined “other race” category.

Decision. Apply the Asian cut-point only where Asian ethnicity can actually be identified — and say so.

Consequence. **At most 17** participants could be grouped differently. An upper bound, not a count of known errors.

Ethnicity variable: RIDRETH3, available 2011–2018 only

The outcome lives in a different dataset

NHANES measures the exposure. It does **not** record who died.

Mortality comes from **NCHS linked mortality files** — separate downloads, joined on SEQN, giving vital status, cause category, and follow-up through **31 Dec 2019**.

◆ **Decision: when does the clock start?** We use time from the **examination**, because that is when BMI, waist and the blood work were measured. The paper does not say which it used.

Warning

A one-column error when reading these fixed-width files corrupts every result **silently** — every model still runs. Sanity-check the parsed values before trusting anything.

Most of it reproduces — that is not the interesting part

	Reproduced	Published
Analytic N	6,371	6,300
Deaths	586	585
Group IV HR, unadjusted	15.14	15.13
Group IV HR, adjusted	2.91	2.89

The interesting part is what the numbers support. Adjusted, Group III is 3.03 and Group IV is 2.89, both against Group I.

*Both groups have higher adjusted mortality than Group I. **Whether Group IV is worse than Group III is a contrast the paper never reports** — and similar point estimates are not evidence of equivalence.*

What to take to P1

For every step of your own paper, write four lines:

Paper — the exact sentence · **Problem** — what it fails to determine ·
Decision — your rule and why · **Consequence** — the change, and its type

Then label each consequence: **eligibility** · **design** · **item missingness** ·
target population · **outcome ascertainment**.



An unexplained exact match is worth less than a discrepancy you can account for. Aim for a **defensible audit trail**, not a matching number.

Where things live: these slides = the reasoning · COHORT_BUILD_STEPS.md = exactly what we did · VARIABLE_MAP.md = where each variable lives · the R/ scripts = proof.

Appendix

Appendix — the full analytic funnel

Step	Criterion	N	Type
0	All NHANES 2007–2018 records	59,842	—
1	Age ≥ 18	36,580	eligibility
2	FLI computable	14,989	design + missingness, fused
3	Fatty liver (FLI ≥ 60)	6,376	target population
4	MASLD (≥ 1 cardiometabolic criterion)	6,374	target population
5	Phenotype group assignable	6,374	missingness
6	Mortality-eligible and linked	6,371	outcome ascertainment
7	Alcohol exclusion applied = 54,655	6,371	analysis setting

Appendix — the fasting-weight domain

Step	Criterion	N	Type
1	Age ≥ 18	36,580	eligibility
2	Positive WTSAF2YR	14,962	design restriction
3	FLI computable	14,170	item missingness
4	Fatty liver (FLI ≥ 60)	6,052	target population
5	MASLD	6,050	target population
7	Mortality- eligible and linked	6,048	outcome ascertainment

6,371 and 6,048 answer different questions. The paper does not report using survey weights, so 6,371 is what reproduces it; 6,048 is the only set on which a design-aware estimate can be computed.

Appendix — “outside the frame” $\neq$ “never asked to fast”

A missing or zero WTSAF2YR conflates at least three different states:

	Adults
No positive fasting weight	21,618
...of whom carry a weight of exactly 0	2,031
...of whom have a triglyceride value anyway	966
...of whom have all four FLI inputs	819

So the weight marks *the analysable fasting domain*. It does not certify who was or was not asked to fast — 819 people have the fasting measurement and no usable weight.

Appendix — the Fatty Liver Index

$$\text{FLI} = \frac{\exp(L)}{1 + \exp(L)} \times 100$$

$$L = 0.953 \cdot \ln(\text{TG}) + 0.139 \cdot \text{BMI} + 0.718 \cdot \ln(\text{GGT}) + 0.053 \cdot \text{waist} - 15.745$$

Input	NHANES variable	Unit
Triglycerides	LBXTR	mg/dL
BMI	BMXBMI	kg/m ²
GGT	LBXSGTSI	U/L
Waist circumference	BMXWAIST	cm

All four arrive in the units the equation expects, so nothing is converted.

Steatosis: **FLI** $\geq$ **60**.

Appendix — the five cardiometabolic criteria

At least one must accompany fatty liver:

1. BMI ≥ 25 , or waist > 94 cm (men) / > 80 cm (women)
2. Fasting glucose ≥ 5.6 mmol/L, HbA1c $\geq 5.7\%$, or antidiabetic treatment
3. Blood pressure $\geq 130/85$ mm Hg, or antihypertensive treatment
4. Triglycerides ≥ 150 mg/dL, or lipid-lowering treatment
5. HDL ≤ 40 mg/dL (men) / < 50 mg/dL (women), or HDL-targeted treatment

Glucose note. 5.6 mmol/L is 100.9 mg/dL; we coded ≥ 100 . 212 people sit in that gap, and re-running MASLD at ≥ 100.9 changes **0** classifications.

Criterion 5 note. NHANES has no variable for HDL-targeted treatment, so that clause is not implementable.

Appendix — the decisions ledger

	Decision or limitation	Consequence
[!]	Age ≥ 18 vs > 18	70 people; evidence points both ways
[!]	Alcohol exclusion built but not applied	427 people if switched on
[!]	cv_death in the saved file = heart + cerebrovascular	185 events vs the paper's 158 (heart only)
◆	Asian cut-point only where ethnicity is identifiable	≤ 17 grouped differently
◆	Full file (6,371) vs fasting-weight domain (6,048)	323 records
◆	Follow-up clock from examination, not interview	mean 6.76 y vs 6.83 y
◆	Composite comorbidity definitions	1–3 percentage-point gaps vs Table 1
◆	No survey weights, matching what the paper reports	the design-aware estimate is P4

Appendix — where each layer lives

Layer	Document	Answers
Reasoning	these slides	what problem are we solving, and why does the choice matter?
Implementation	COHORT_BUILD_STEPS.md	exactly what did we do, step by step, with every number?
Lookup	VARIABLE_MAP.md	where does this variable live, which cycles have it, what do its codes mean?
Proof	R/00-07	run it yourself

The full audit version of this deck — every line reference, every count — is Cohort_recipe_slides.qmd.